

Remarce Williams (DSW9740)

(1) (a). D-H Table

Link	a_i	d_i	α_i	θ_i
1	a_1	0	0	θ_1
2	a_2	0	0	θ_2
3	0	d_3	0	0
4	0	0	0	θ_4

$$(b) T_1^0 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 = \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_4^3 = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(c) T_4^0 = T_1^0 \cdot T_2^1 \cdot T_3^2 \cdot T_4^3$$

$$\begin{bmatrix} c_1 & -s_1 & 0 & a_1 c_1 \\ s_1 & c_1 & 0 & a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_2 & -s_2 & 0 & a_2 c_2 \\ s_2 & c_2 & 0 & a_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} c_{12} & -s_{12} & 0 & a_2 c_{12} + a_1 c_1 \\ s_{12} & c_{12} & 0 & a_2 s_{12} + a_1 s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

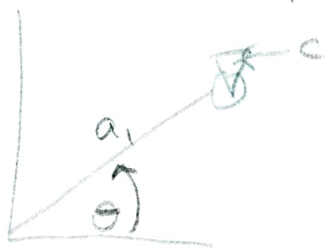
$$= \begin{bmatrix} c_{12}c_4 - s_{12}s_4 & -c_{12}s_4 - s_{12}c_4 & 0 & a_1c_1 + a_2c_{12} \\ s_{12}c_4 + c_{12}s_4 & -s_{12}s_4 + c_{12}c_4 & 0 & a_1s_1 + a_2s_{12} \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(2) If C is parallel to e
 $\Rightarrow$ they have the same orientation

$$\text{If } T_e^0(q) = \begin{bmatrix} 0 & s_{12} & c_{12} & a_1 c_1 + a_2 c_{12} \\ 0 & -c_{12} & s_{12} & a_1 s_1 + a_2 s_{12} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_c^0 = \begin{bmatrix} 0 & s_{12} & c_{12} \\ 0 & -c_{12} & s_{12} \\ 1 & 0 & 0 \end{bmatrix}$$

using trigonometry



$$P_c^0 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore T_c^0 = \begin{bmatrix} R_c^0 & P_c^0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & s_{12} & c_{12} & a_1 c_1 \\ 0 & -c_{12} & s_{12} & a_1 s_1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{if } p^c = \begin{bmatrix} 10 \\ -10 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} p^b \\ 1 \end{bmatrix} &= \frac{b}{c} \cdot \begin{bmatrix} p^c \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & s_{12} & c_{12} & a_1 c_1 \\ 0 & -c_{12} & s_{12} & a_1 s_1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ -10 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} a_1 c_1 - 10 s_{12} \\ a_1 s_1 + 10 c_{12} \\ 10 \\ 1 \end{bmatrix} \end{aligned}$$

$$\therefore p^b = \begin{bmatrix} a_1 c_1 - 10 s_{12} \\ a_1 s_1 + 10 c_{12} \\ 10 \end{bmatrix}$$